

SUWA, Japan

WMO: 476200

Lat: 36.0461N

Lon: 138.1097E

Elev: 2502

StdP: 13.41

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	18.4	7.1	8.7	24.1	9.5	9.8	25.0	23.0	32.2	20.9	32.3	3.8	160	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	88.6	72.7	86.3	71.9	83.9	71.1	74.5	84.4	73.5	82.6	72.5	80.8	8.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	128.9	79.1	70.8	124.8	78.1	69.9	120.9	77.1	40.0	85.0	38.9	83.0	38.0	81.0	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	19.0	16.7	10.9	91.5	2.7	1.8	8.9	92.9	7.3	94.0	5.8	95.0	3.8	96.3
			9.8	76.8	2.6	1.3	8.0	77.7	6.5	78.5	5.0	79.2	3.2	80.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	30.0	31.8	39.5	50.4	60.5	67.4	75.0	76.7	68.6	57.2	45.6	35.8
	DBStd	16.86	4.72	5.75	6.01	6.61	4.96	4.27	4.16	3.72	5.66	5.55	5.47	5.24
	HDD50	2136	620	510	329	74	2	0	0	0	0	7	153	441
	HDD65	5188	1085	930	791	439	155	24	0	0	30	248	581	905
	CDD50	3350	0	0	3	86	326	523	775	826	558	231	22	0
	CDD65	926	0	0	0	0	14	97	310	361	137	7	0	0
	CDH74	5527	0	0	0	9	128	416	1952	2436	573	13	0	0
	CDH80	1553	0	0	0	0	11	45	601	795	101	0	0	0
Wind	WSAvg	6.8	6.7	7.3	8.3	8.1	7.4	6.3	5.8	6.0	6.4	6.5	6.4	6.5
Precipitation	PrecAvg	45.6	1.7	1.9	3.1	3.3	4.1	5.3	6.1	4.7	6.6	4.6	2.7	1.5
	PrecMax	59.1	4.4	3.9	5.7	7.7	9.6	11.0	16.9	8.9	13.5	14.6	6.8	3.1
	PrecMin	29.3	0.1	0.6	1.1	0.9	1.8	2.2	0.7	1.5	1.5	0.2	0.3	0.0
	PrecStd	8.3	1.1	1.0	1.2	1.6	1.9	1.9	3.5	2.2	2.9	2.9	1.7	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	54.7	63.7	74.7	81.5	84.3	91.0	91.7	86.5	75.8	64.6	55.1
		MCWB	40.3	45.6	47.8	56.2	60.8	66.7	73.3	73.2	71.5	65.2	54.6	47.5
	2%	DB	44.2	48.9	58.8	69.9	77.7	80.9	88.6	89.1	82.9	71.9	60.9	50.5
		MCWB	37.6	40.7	45.8	54.0	59.8	65.9	73.0	72.6	70.0	62.3	52.7	44.7
	5%	DB	41.0	44.8	54.6	66.1	74.3	78.5	86.2	87.0	80.0	69.0	57.9	47.9
		MCWB	35.1	37.7	44.2	52.0	58.5	65.0	72.1	72.3	69.0	60.7	51.3	42.2
	10%	DB	38.4	41.5	51.0	62.7	71.2	76.3	83.5	84.7	77.2	66.3	55.2	44.9
		MCWB	33.1	35.8	43.0	51.3	57.5	64.7	71.2	71.9	67.9	59.2	49.9	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	46.8	51.7	62.0	66.5	71.9	76.2	76.1	73.6	67.2	58.2	50.4
		MCDB	45.3	51.8	56.0	65.6	72.9	78.8	87.2	86.9	82.4	71.7	61.2	52.3
	2%	WB	38.5	42.0	48.8	57.9	63.8	70.0	74.5	74.7	71.7	64.8	55.2	46.3
		MCDB	42.6	46.9	54.9	65.0	70.4	75.6	84.5	84.7	79.3	69.9	58.3	49.8
	5%	WB	35.8	38.9	46.4	55.2	61.6	68.6	73.4	73.8	70.5	62.5	52.8	42.7
		MCDB	39.9	42.9	52.2	62.5	70.0	74.4	82.6	83.1	77.2	67.4	56.4	46.5
	10%	WB	33.8	36.5	43.8	52.9	59.7	67.0	72.3	72.9	69.4	60.2	50.5	40.2
		MCDB	37.5	40.9	50.1	60.6	68.1	73.2	80.8	81.4	75.3	65.0	54.5	44.2
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.7	18.0	19.6	18.8	15.4	14.2	15.0	14.0	15.3	16.3	15.0
		MCDBR	18.5	21.6	24.5	25.3	24.4	19.4	18.3	18.4	16.6	17.6	19.7	18.5
		MCWBR	12.6	14.0	14.3	12.5	10.8	7.9	6.7	6.4	6.9	9.4	12.3	12.9
	5% WB	MCDBR	16.3	17.8	20.5	19.8	19.3	14.3	16.6	16.3	14.3	14.2	15.8	16.1
		MCWBR	12.0	12.9	14.3	11.6	10.3	7.1	6.7	6.4	6.5	8.5	11.3	12.5
Clear-Sky Solar Irradiance	taub	0.346	0.388	0.456	0.503	0.523	0.559	0.542	0.549	0.474	0.423	0.369	0.342	
	taud	2.166	2.054	1.933	1.887	1.895	1.871	1.985	1.958	2.119	2.180	2.269	2.263	
	Ebn at Noon	262	263	255	250	247	238	241	236	248	249	253	256	
	Edh at Noon	36	45	56	62	62	64	56	57	46	40	32	31	
All-Sky Solar Radiation	RadAvg	751	984	1258	1504	1664	1497	1490	1506	1218	1011	788	662	
	RadStd	48	64	85	132	173	110	213	143	104	99	63	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page